

### Task 3: Bandit-specific Metrics (A–I)

#### A. Personalization Gain (PG)

- Measures per-user improvement of the bandit policy vs a fixed baseline (e.g., Pomodoro).
- Computation:  $PG_u = \text{avg reward under bandit} - \text{avg reward under baseline}$  (using microrandom probes or baseline sessions).
- Interpretation: Positive PG shows personalization benefit for that user.
- Validation: Use MovieLens simulated bandit or microrandomized probes.

#### B. Regret-per-Hour (RPH)

- Measures cumulative regret normalized by session time.
- Computation:  $RPH = (\Sigma(r^* - r)) / \text{total\_session\_hours}$ .
- Interpretation: Low RPH indicates fast learning and good experience.
- Validation: Simulated bandit environments with oracle baseline.

#### C. Adaptation Half-Life (AHL)

- Measures how quickly the bandit recovers after context distribution shift.
- Computation: Time until reward reaches midpoint between pre- and post-shift steady state.
- Interpretation: Shorter AHL = faster adaptation.
- Validation: Synthetic nonstationary simulations (RecoGym / custom).

#### D. Exploration Overhead Index (EOI)

- Measures the cost of exploration on short-term reward.
- Computation:  $EOI = \text{avg reward during explore} - \text{avg reward during exploit}$  (or loss version).
- Interpretation: Quantifies user cost of learning.
- Validation: Logged bandit datasets with explore/exploit flags (e.g., Open Bandit).

#### E. Break Utility Curve (BUC) & AUC-BUC

- Measures dose–response curve of break length → immediate benefit.
- Computation: Aggregate post-break uplift by break length, integrate curve for AUC-BUC.

- Interpretation: Identifies sweet-spot break length and overall benefit.
- Validation: Field logs with varied break lengths or microrandom probes.

#### **F. Load-Recovery Slope (LRS)**

- Measures recovery speed of load after a break.
- Computation: Fit slope of load vs. time in first N minutes post-break;  $LRS = -\text{slope}$ .
- Interpretation: Larger LRS = faster recovery.
- Validation: On-device load estimator outputs.

#### **G. Counterfactual Targeting Uplift (CTU)**

- Measures causal uplift of break decisions in chosen contexts.
- Computation:  $CTU(x) = E[Y|do(break),x] - E[Y|do(no\ break),x]$  via IPS/DR estimators.
- Interpretation: Checks if breaks are targeted where they help most.
- Validation: Logged data with propensities; Open Bandit or own MRT data.

#### **H. Stability–Productivity Frontier (SPF)**

- Measures trade-off between mean reward and variance across policy settings.
- Computation: Plot mean vs variance across hyperparameters to get frontier.
- Interpretation: Evaluates robustness vs performance.
- Validation: Logged data or simulations with multiple policy variants.

#### **I. Safety-Violation Rate (SVR)**

- Measures how often safety constraints override bandit decisions.
- Computation:  $SVR = (\# \text{ safety overrides}) / (\# \text{ total decisions})$ .
- Interpretation: Should be near zero; important for ethics & compliance.
- Validation: Field logs with guardrail tracking.